

Chapter - 1

Sexual Reproduction in Flowering Plants

NEET UG
Biology

Introduction :

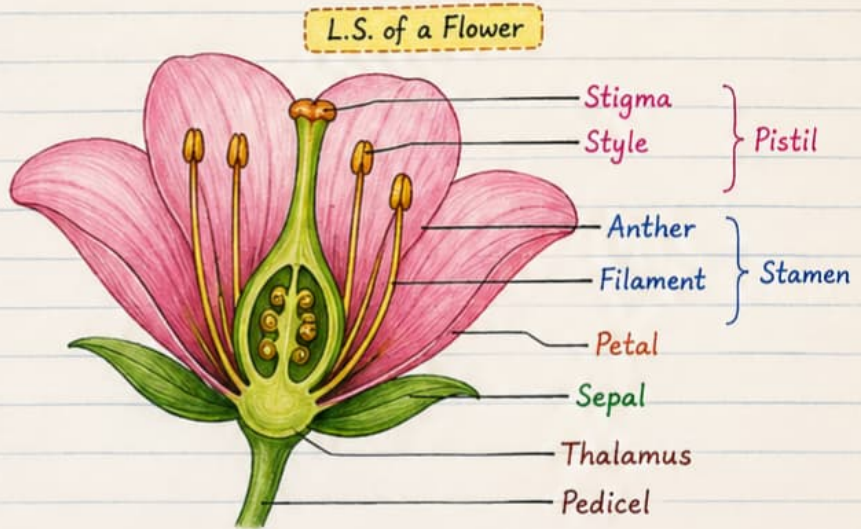
- ★ Flowering plants (Angiosperms) show sexual reproduction.
- ★ It involves the fusion of male and female gametes to form a **zygote**.
- ★ This process occurs in the flower.
- ★ It leads to the formation of **seeds** and **fruits**.

Key Point

Sexual reproduction
↑ Genetic variation
↓
Better adaptation
and evolution

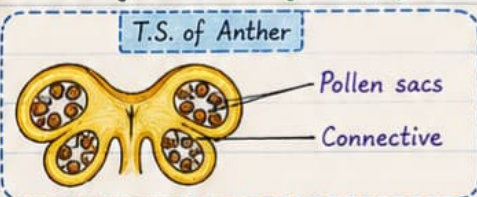
Parts of a Flower :

- ① **Calyx** – Outer most whorl, formed by sepals.
- ② **Corolla** – Next whorl, formed by petals.
- ③ **Androecium** – Male reproductive part (stamen).
- ④ **Gynoecium** – Female reproductive part (pistil).
- ⑤ **Thalamus** – Receptacle on which all parts are borne.



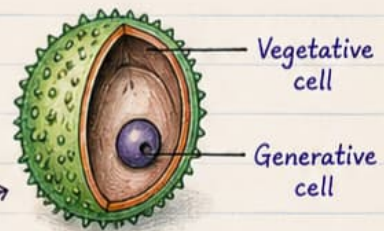
Androecium (Male part) :

- Each stamen has two parts – **Anther** and **Filament**.
- Anther is bilobed, produces **pollen grains**.
- Pollen grain = **Male gametophyte**.



Microspore (2n)
↓ Meiosis
Microspores (n)
↓
Pollen grains (n)

Each pollen grain contains :
• Vegetative cell
• Generative cell



Gynoecium (Female part) :

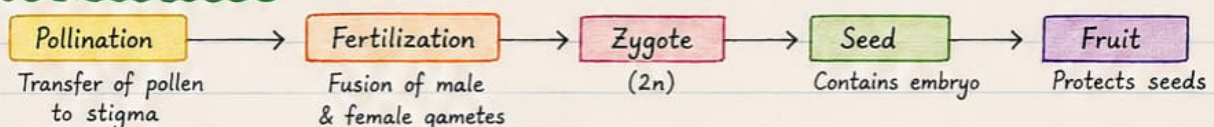
- Made up of one or more carpels.
- Each carpel has 3 parts – **Stigma**, **Style**, **Ovary**.
- Ovary contains one or more **ovules**.
- Ovule contains **female gametophyte (embryo sac)**.

L.S. of Pistil



Megaspore mother cell (2n)
↓ Meiosis
Megaspore (n)
↓
Embryo sac (female gametophyte)

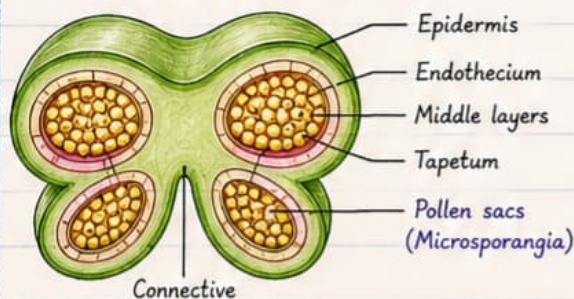
Process Overview :



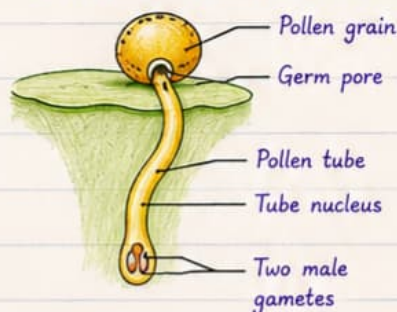
Sexual Reproduction in Flowering Plants

Pollination - Transfer of pollen grains from an anther to stigma.

T.S. of 'Anther (Young)



Pollen Germination on Stigma



Note

Pollen tube grows through style and reaches the ovule in the ovary.

Types of Pollination :

1. Self Pollination (Autogamy)

- Pollen transfer occurs from anther to stigma of the **same flower** or another flower on the **same plant**.
- Does not require any external agent.
- Example - **Pea, Wheat, Rice, Tomato**.



Pollen from anther reaches stigma of the same flower

2. Cross Pollination (Allogamy)

- Pollen transfer occurs from anther of one flower to stigma of another flower on a **different plant** of the same species.
- Requires external agents.
- Example - **Maize, Papaya, Mango**.



Pollen from anther of one flower reaches stigma of another flower

Agents of Pollination

1. Wind (Anemophily)

- Flowers are small, inconspicuous.
- Pollen grains are light, dry, non-sticky.
- Produced in large quantity.
- Example - **Grass, Maize**.



2. Water (Hydrophily)

- Pollen grains are carried by water.
- Flowers are small and inconspicuous.
- Example - **Vallisneria, Zostera**.



3. Insects (Entomophily)

- Flowers are large, colourful, scented.
- Pollen grains are sticky, heavy.
- Nectar is present.
- Example - **Hibiscus, Sunflower**.

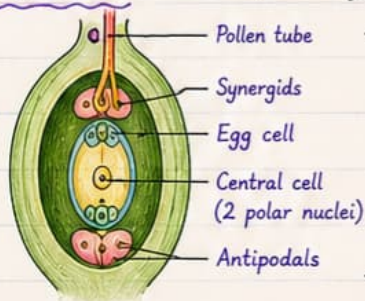


4. Birds (Ornithophily)

- Flowers are tubular, bright coloured.
- Nectar is present.
- Pollen grains are sticky.
- Example - **Butea, Erythrina**.



Fertilization - Fusion of male gamete with female gamete.



(Double Fertilization)

- ① One male gamete + Egg cell (n)
→ **Zygote (2n)**
→ Develops into **Embryo**
- ② Second male gamete + 2 Polar nuclei (n+n)
→ **Primary endosperm nucleus (3n)**
→ Develops into **Endosperm**

Significance

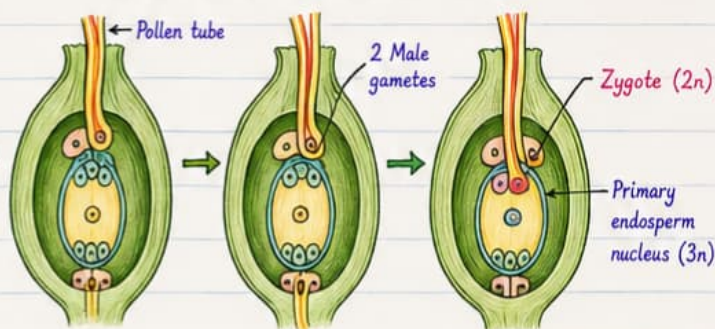
- ✓ Ensures formation of viable seeds.
- ✓ Maintains genetic variation.
- ✓ Helps in adaptation and evolution.

★ **Key Takeaway :** Sexual reproduction in flowering plants leads to formation of **seeds** and **fruits** through pollination, fertilization and double fertilization.

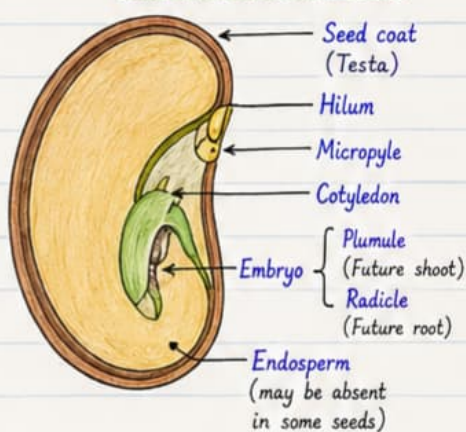
Sexual Reproduction in Flowering Plants

Double Fertilization – Unique feature of Angiosperms.

- Discovered by – Nawaschin (1898).
- Pollen tube enters the ovule through **micropyle** and releases two male gametes into the embryo sac.
- One male gamete fuses with egg cell → forms **zygote (2n)**.
- Other male gamete fuses with two polar nuclei → forms **primary endosperm nucleus (3n)**.
- This process is called **double fertilization**.

Steps of Double FertilizationAfter Fertilization :

Part	Develops into
Zygote (2n)	Embryo
Primary endosperm nucleus (3n)	Endosperm (food for embryo)
Ovule	Seed
Integuments	Seed coat
Ovary	Fruit
Ovary wall	Pericarp (fruit wall)
Micropyle	Hilum (scar on seed)

Mature Seed (T.S.)Note

In dicot seeds (e.g., Pea, Gram), food is stored in cotyledons.

In monocot seeds (e.g., Maize, Wheat), food is stored in endosperm.

Fruit – Matured ovary► Functions –

- ✓ Protects the seeds.
- ✓ Helps in seed dispersal.
- ✓ Ensures development of seeds.

Types of Fruits1. Simple Fruits – Develop from a single ovary.

- Example – Mango, Tomato, Pea.

2. Aggregate Fruits – Develop from a single flower having many ovaries (apocarpous).

- Example – Custard apple, Raspberry.

3. Multiple Fruits – Develop from an inflorescence.

- Example – Pineapple, Jackfruit.

Modes of Seed DispersalWindWaterAnimalsExplosionQuick Revision – Key Points

- ★ Sexual reproduction involves fusion of male and female gametes.
- ★ Flower is the reproductive unit of angiosperms.
- ★ Pollination brings pollen to the stigma.
- ★ Fertilization leads to double fertilization.
- ★ Zygote → Embryo (2n), Primary endosperm nucleus → Endosperm (3n).
- ★ Ovule → Seed and Ovary → Fruit.
- ★ Fruits protect seeds and help in their dispersal.

★ Remember!

“Double fertilization is the most outstanding event in the life cycle of flowering plants.”



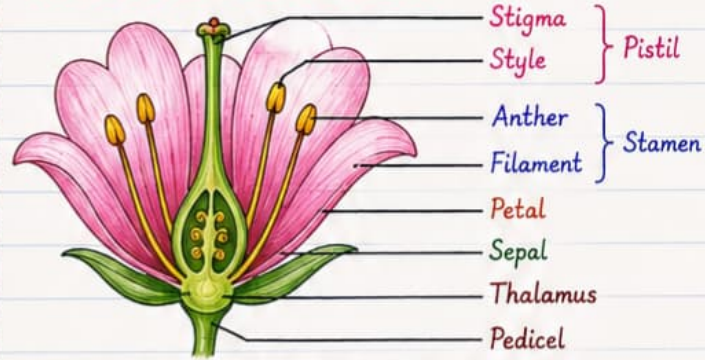
Sexual Reproduction in Flowering Plants

Significance of Sexual Reproduction

- ★ Creates **genetic variation**.
- ★ Helps in better adaptation to environment.
- ★ Leads to **evolution** of new species.
- ★ Ensures continuity of species.



Structure of a Typical Flower



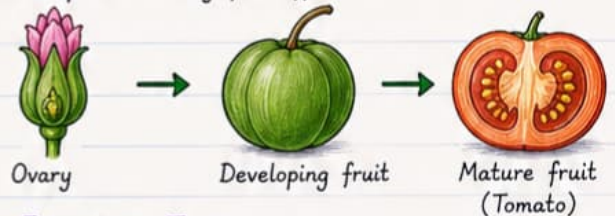
Complete flower = Both **Stamen** and **Pistil** present.
Incomplete flower = Either **Stamen** or **Pistil** absent.

Differences : Self vs Cross Pollination

Feature	Self Pollination	Cross Pollination
Pollen source	Same flower or same plant	Different plant of same species
Genetic variation	Very less	High
Offspring	More similar to parent	More variation and vigor
Adaptability	Low	High
Agents	Not required	Wind, Water, Insects, etc.

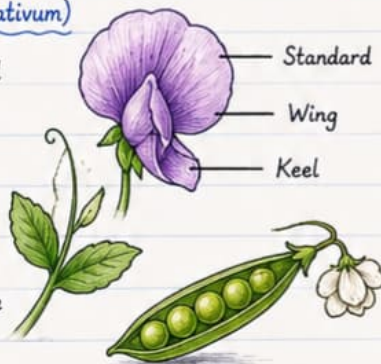
Post Fertilization Changes in Flower

- **Petals, sepals, stamens** → Wither and fall off.
- **Ovary** → Develops into **fruit**.
- **Ovules** → Develops into **seeds**.
- **Ovary wall** → Develops into **fruit wall** (pericarp).
- **Other parts** → May fall off.



Example - Pea (*Pisum sativum*)

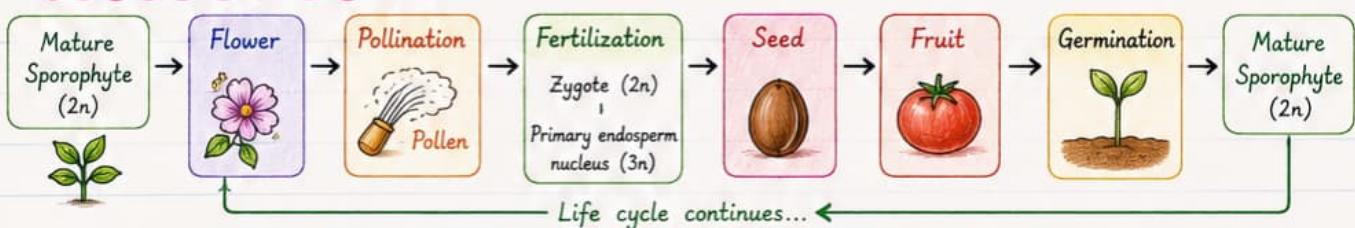
- Flowers are bisexual and usually **self-pollinated**.
- **Cleistogamous flowers** → Always self-pollinated (do not open).
- **Chasmogamous flowers** → May self or cross pollinate (open flowers).



Important Terms

- ★ **Gamete** - Male or female sex cell.
- ★ **Zygote** - Formed by fusion of male and female gametes ($2n$).
- ★ **Embryo** - Young sporophyte developed from zygote.
- ★ **Endosperm** - Nutritive tissue for embryo ($3n$ in angiosperms).
- ★ **Seed** - Mature ovule containing embryo, endosperm and seed coat.
- ★ **Fruit** - Mature ovary containing seeds.

Life Cycle of a Flowering Plant



One Line Summary

Pollination brings pollen to **stigma**, fertilization forms **zygote** and **endosperm**, ovule becomes **seed**, ovary becomes **fruit** - ensuring continuation of life cycle.